53

Urinary Tract Infections in Adults

Dimitri Drekonja, Thomas Hooton

DEFINITION

Most episodes of urinary tract infection (UTI) in adults can be categorized into six common categories: females with acute uncomplicated cystitis, females with recurrent cystitis, females and males with acute uncomplicated pyelonephritis, males with cystitis, complicated UTI, and asymptomatic bacteriuria (Box 53.1).[1] Additionally, UTIs in chronic kidney disease (CKD), those needing urethral catheterization or other indwelling devices, and patients with a spinal cord injury require different approaches to diagnosis and management. Chapter 44 discusses UTI in pregnancy, and Chapter 62 describes vesicoureteral reflux (VUR) in children. UTI in autosomal dominant polycystic kidney disease is discussed in Chapter 46.

Complicated UTI is defined as UTI occurring in a patient with a condition that increases the risk for treatment failure or recurrence. Potentially complicating conditions are presented in Box 53.2; not all of these conditions confer a similar risk for treatment failure or recurrence. Complicated UTIs may require different pretreatment and posttreatment evaluation than do uncomplicated UTIs, and type and duration of antimicrobial treatment may also differ. On occasion, complicated UTIs are diagnosed only after a patient has a poor response to treatment and the underlying condition is identified.

EPIDEMIOLOGY

Several million episodes of acute uncomplicated cystitis and at least 250,000 episodes of acute pyelonephritis occur annually in the United States. The incidence of cystitis in sexually active young females is about 0.5 per 1 person-year.[2] Acute uncomplicated cystitis may recur in 27% to 44% of healthy females.[3] The incidence of pyelonephritis in young females is about 3 per 1000 person-years.[4] The self-reported incidence of symptomatic UTI in postmenopausal females is about 10% per year.[5] The incidence of symptomatic UTI in adult males younger than 50 years is lower than in females, ranging from 5 to 8 per 10,000 males annually.

Complicated UTIs occur in a wide range of settings (see Box 53.2). Nosocomial UTIs are a common type of complicated UTI and occur in 5% of admissions in the university tertiary care hospital setting; catheter-associated infections account for most of the infections. Catheter-associated bacteriuria is the most common source of gram-negative bacteremia in hospitalized patients.[6]

Asymptomatic bacteriuria is defined as the presence of two separate consecutive clean-voided urine specimens, both with 10^5 or more colony-forming units per milliliter (cfu/mL) of the same uropathogen in the absence of symptoms referable to the urinary tract.[7] Asymptomatic bacteriuria occurs in 5% of young adult females, but rarely in males younger than 50 years. The prevalence increases to 16% of ambulatory females and 19% of ambulatory males older than 70 and up to 50% of elderly females and 40% of elderly males who are institutionalized.[7] In clinical practice, asymptomatic bacteriuria is rarely confirmed by two consecutive specimens. Asymptomatic bacteriuria may be persistent or transient and recurrent, and many patients have had previous symptomatic infection or develop symptomatic UTI soon after having asymptomatic bacteriuria. Asymptomatic bacteriuria is generally benign but occasionally may lead to serious complications.

PATHOGENESIS

Uncomplicated Infection

Most uncomplicated UTIs in healthy females begin with uropathogens (typically *Escherichia coli*) colonizing the rectal flora entering the urethra and then the bladder, after an interim phase of introital colonization. Colonizing uropathogens also may come from a sex partner's vagina, rectum, or penis. Hematogenous seeding of the urinary tract by potential uropathogens such as *Staphylococcus aureus* is the source of some UTIs, but this is more likely to occur in the setting of persistent bloodstream infection or urinary tract obstruction.

Many host behavioral, genetic, and biologic factors predispose healthy young females to uncomplicated UTI (Table 53.1).[2,8] Factors protecting individuals from UTI include the host's immune response; maintenance of normal vaginal flora, which protects against colonization with uropathogens; and removal/prevention of bladder bacteriuria by micturition.[9] Uropathogenic *E. coli,* the predominant pathogens in uncomplicated UTI, are a subset of extraintestinal pathogenic *E. coli* that have enhanced virulence[10] (see Table 53.1). P-fimbriated strains of *E. coli* are associated with acute uncomplicated pyelonephritis, and their adherence properties may stimulate epithelial and other cells to produce proinflammatory factors that stimulate the inflammatory response.[11] Other virulence determinants include adherence factors (type 1, S, and Dr fimbriae), toxins (hemolysin), immune evasion, iron acquisition (aerobactin), flagella, and serum resistance.[10,12] Bacterial determinants associated with cystitis and

BOX 53.1 Categories of Urinary Tract Infection in Adults

- Acute cystitis in females
- Recurrent cystitis in females
- Acute pyelonephritis in females and males
- Acute cystitis in males
- Complicated urinary tract infection[a]
- Asymptomatic bacteriuria

[a]This is a selected list of complicating factors. Some factors complicate urinary tract infections through several mechanisms.
Data from Nicolle LE. A practical guide to the management of complicated urinary tract infection. *Drugs*.1997;53:583–592.

BOX 53.2 Selected List of Potential Complicating Factors for Urinary Tract Infections

- Male sex
- Pregnancy
- Poorly controlled diabetes mellitus
- Obstruction or other structural factor: Urolithiasis, malignancies, ureteral and urethral strictures, bladder diverticula, renal cysts, fistulas, ileal conduits, other urinary diversions
- Functional abnormality: Neurogenic bladder, vesicoureteral reflux
- Foreign bodies: Indwelling catheter, ureteral stent, nephrostomy tube
- Other conditions: kidney failure, kidney transplantation, immunosuppression, multidrug-resistant uropathogens, health care–associated (includes hospital-acquired long-term care facility acquired infection, prostatitis-related infection, upper tract infection in an adult other than a healthy female, other functional or anatomic abnormality of urinary tract)

Data from Nicolle LE. A practical guide to the management of complicated urinary tract infection. *Drugs*. 1997;53:583-592.

asymptomatic bacteriuria are less characterized, and triggers for urinary symptoms are unclear.

Factors affecting the large difference in UTI prevalence between males and females include the greater distance between the anus and the urethral meatus in males, the drier environment surrounding the male urethra, and the greater length of the male urethra. Risk factors associated with UTIs in healthy males include intercourse with a female partner colonized or infected with a potential uropathogen, anal intercourse, and being uncircumcised, although these factors are often not present in males with UTI. Most uropathogenic strains infecting young males are highly virulent, suggesting that the urinary tract in healthy males is relatively resistant to infection.

Complicated Infection

The initial steps leading to uncomplicated UTI discussed earlier probably also occur in most individuals who develop a complicated UTI. Factors that predispose individuals to complicated UTI generally do so by causing obstruction or stasis of urine flow, facilitating entry of uropathogens into the urinary tract by bypassing normal host defense mechanisms, providing a nidus for infection that is not readily treatable with antimicrobials, or compromising the host immune system (see Box 53.1).[1] UTIs are more likely to become complicated if host defense is impaired, as occurs with indwelling catheter use, VUR, obstruction, neutropenia, and immune deficiencies. Diabetes mellitus is associated with several UTI syndromes, including renal and perirenal abscesses, emphysematous pyelonephritis and cystitis, papillary necrosis, and xanthogranulomatous pyelonephritis.[12] Uropathogen

TABLE 53.1 Factors Modulating Risk for Acute Uncomplicated Urinary Tract Infections in Females

Host Determinants	Uropathogen Determinants
Behavioral: Sexual intercourse, use of spermicidal products, recent antimicrobial use, suboptimal voiding habits *Genetic:* Innate and adaptive immune response, enhanced epithelial cell adherence, antibacterial factors in urine and bladder mucosa, nonsecretor of ABO blood group antigens, P1 blood group phenotype, reduced *CXCR1* expression, previous history of recurrent cystitis *Biologic:* Estrogen deficiency in postmenopausal females, glycosuria (including from SGLT-2 inhibitors)	*Escherichia coli* virulence determinants: P, S, Dr, and type 1 fimbriae; hemolysin; aerobactin; serum resistance

virulence determinants are less important in the pathogenesis of complicated UTIs compared with uncomplicated UTIs. However, infection with multidrug-resistant uropathogens is more likely with complicated UTI, and the causative organisms are also more varied.

ETIOLOGIC AGENTS

Uncomplicated upper and lower UTI are caused by *E. coli*, present in 70% to 95% of infections, and *Staphylococcus saprophyticus*, present in 5% to more than 20%. Other organisms are less common (Table 53.2).[7] *S. saprophyticus* only rarely causes acute pyelonephritis.[13] Among healthy nonpregnant females, the isolation of lactobacilli, enterococci, group B streptococci, and coagulase-negative staphylococci other than *S. saprophyticus* usually represents contamination of the urine specimen[14] unless found in voided midstream urine in high counts and pure growth in symptomatic females.

A broader range of bacteria can cause complicated UTI, many demonstrating resistance to multiple antimicrobial agents. Although *E. coli* is most commonly isolated, *Citrobacter* spp., *Enterobacter* spp., *Pseudomonas aeruginosa*, enterococci, and *S. aureus* account for a higher proportion of cases relative to uncomplicated UTIs (see Table 53.2).[1] The proportion of infections caused by fungi, especially *Candida* spp., is increasing (see Chapter 55). Patients with chronic conditions, such as spinal cord injury and neurogenic bladder, are more likely to have polymicrobial and multidrug-resistant infections.

CLINICAL SYNDROMES

Acute Uncomplicated Cystitis in Young Females

Females with acute uncomplicated cystitis generally present with dysuria, frequency, urgency, or suprapubic pain. The differential diagnosis for acute dysuria in a sexually active young female is acute cystitis; acute urethritis from *Chlamydia trachomatis, Neisseria gonorrhoeae*, or herpes simplex virus infections; or vaginitis caused by *Candida* spp. or *Trichomonas vaginalis*.[6] These three entities can usually be distinguished by the history, physical examination, and simple laboratory tests. New vaginal discharge supports urethritis or vaginitis and should prompt a vaginal exam, whereas its absence is more consistent with

TABLE 53.2 Bacterial Etiology of Urinary Tract Infections

	URINARY TRACT INFECTION (%)	
Organisms	**Uncomplicated**	**Complicated**
Gram-Negative Organisms		
Escherichia coli	70–95	21–54
Proteus mirabilis	1–2	1–10
Klebsiella spp.	1–2	2–17
Citrobacter spp.	<1	5
Enterobacter spp.	<1	2–10
Pseudomonas aeruginosa	<1	2–19
Other	<1	6–20
Gram-Positive Organisms		
Coagulase-negative staphylococci *(Staphylococcus saprophyticus)*	5–20+	1–4
Enterococci	1–2	1–23
Group B streptococci	<1	1–4
Staphylococcus aureus	<1	1–2
Other	<1	2

Data from Nicolle LE. A practical guide to the management of complicated urinary tract infection. *Drugs*. 1997;53:583–592.

cystitis. Skin lesions may provide a clue that herpes simplex virus is the cause of the dysuria. Pyuria is present in almost all females with acute cystitis and in most females with urethritis caused by *N. gonorrhoeae* or *C. trachomatis,* and its absence strongly suggests an alternative diagnosis. Hematuria (microscopic or gross) is common in females with UTI but not in females with urethritis or vaginitis.

Definitive diagnosis of UTI requires the presence of significant bacteriuria, the traditional standard for which is up to 10^5 uropathogens per milliliter of voided midstream urine. A minority of females with cystitis have lower colony counts, which are missed with use of the traditional definition.[15] Despite their central role in the definitive diagnosis of UTI, urine cultures are generally not indicated in females with uncomplicated cystitis, in particular for females with a prior culture-confirmed UTI. For these females, history is highly reliable in establishing the diagnosis,[16] the causative organisms are predictable, and the culture results usually become available only after therapeutic decisions have been made and treatment is nearly complete.

E. coli isolates from outpatients with UTI are increasingly resistant to commonly used antibiotics, including sulfonamides, amoxicillin, trimethoprim, and trimethoprim-sulfamethoxazole (TMP-SMX).[17–19] In the United States, TMP-SMX resistance among *E. coli* isolates causing uncomplicated UTI range from 15% to 42% in different regions,[17] with a similar range among European countries and Brazil.[18] Many drug-resistant *E. coli* strains are clonal and have been hypothesized to enter new environments by contaminated products ingested by community residents.[20] Because of increasing resistance to TMP-SMX and other antimicrobials, clinicians have turned to agents with more predictable activity. Nitrofurantoin is active against most *E. coli* isolates, with resistance rates of less than 5%. With *E. coli* being the most isolated organism for females with uncomplicated cystitis, nitrofurantoin is a first-line agent for this syndrome. Fluoroquinolones remain active against most *E. coli* strains causing uncomplicated cystitis, although resistance is increasing in most locations and use for cystitis is discouraged.[17,18,21] In an antimicrobial susceptibility study of more than 12 million *E. coli* isolates from U.S. outpatients, fluoroquinolone resistance increased from 3% to 17% over 10 years.[22] In addition, infections caused by extended-spectrum β-lactamase (ESBL)-producing strains are increasingly common, even for uncomplicated cystitis.

Recommended management of acute uncomplicated cystitis is summarized in Fig. 53.1 and Table 53.3. Updated Infectious Diseases Society of America (IDSA) guidelines emphasize the importance of considering ecologic adverse effects of antimicrobial agents (i.e., selection for colonization or infection with multidrug-resistant organisms—so-called collateral damage) when selecting a treatment regimen.[21] Short-course regimens are recommended as first-line treatment for acute uncomplicated cystitis because of comparable efficacy, better compliance, lower cost, and fewer adverse effects than with longer regimens. Given the benign nature of uncomplicated cystitis along with its high frequency, the IDSA guidelines give equal weight to the risk for ecologic adverse effects and drug effectiveness in the recommendations. Nitrofurantoin is well tolerated and has good efficacy when given twice daily for 5 days, and it has a low propensity for ecologic adverse effects.[23] Despite concern about the high prevalence of resistance to TMP-SMX, this combination medication remains effective for infections caused by susceptible isolates and is inexpensive and well tolerated. Fosfomycin is also considered a first-line regimen because of its low propensity for ecologic adverse effects and single-dose administration, even though it appears to be slightly inferior to TMP-SMX and fluoroquinolones, and is often considerably more expensive than other options.[21] Moreover, both nitrofurantoin and fosfomycin appear to have a role as therapeutic agents effective against ESBL *E. coli* UTIs.[24,25] Pivmecillinam, an extended gram-negative spectrum penicillin used only for treatment of UTI, is an appropriate choice for therapy in regions where it is available (primarily European countries) because of minimal resistance and propensity for collateral damage and activity against ESBL-producing organisms.[26] The choice of an antimicrobial agent should be individualized based on the patient's allergy and medication preferences, local practice patterns, prevalence of resistance in the local community (if known), availability, cost, and how comfortable the patient and provider are with increased risk for treatment failure associated with the use of an antimicrobial agent that may cause less collateral damage but is less effective than another drug.[21] If first-line antimicrobial agents are not suitable because of one or more of these factors, fluoroquinolones or β-lactams are reasonable alternatives, although their use should be minimized because of concerns for increasing the risk for drug resistance.[21] Thus, although fluoroquinolones (3-day duration) are highly effective in the treatment of cystitis, many experts recommend that they be considered as second-line therapy for uncomplicated cystitis to help preserve their usefulness in the treatment of other more serious infections.[21] Moreover, in the United States, the Food and Drug Administration (FDA) has stated that the risks of systemic fluoroquinolones outweigh their benefits for uncomplicated cystitis.[27] In general, β-lactam antibiotics have been inferior to TMP-SMX or fluoroquinolones in regimens of the same duration.[28]

Although broad-spectrum oral β-lactams (e.g., cefixime, cefpodoxime, cefprozil, cefaclor, amoxicillin-clavulanate) demonstrate in vitro activity against most uropathogens causing uncomplicated cystitis, clinical data are sparse. Trials have shown that cure rates with 3-day regimens of amoxicillin-clavulanate[29] or cefpodoxime proxetil[28] were lower than a 3-day regimen of ciprofloxacin. Moreover, there are concerns about the possibility of ecologic adverse effects with oral broad-spectrum cephalosporins, as has been observed with parenteral cephalosporins. Routine posttreatment cultures in females are not indicated unless the patient is symptomatic. If the patient remains symptomatic and persistent infection is confirmed, a longer course of culture-directed therapy (usually with a fluoroquinolone) should be used. Detecting and treating asymptomatic bacteriuria in healthy females is useful only in pregnancy and before urologic instrumentation or surgery.[7]

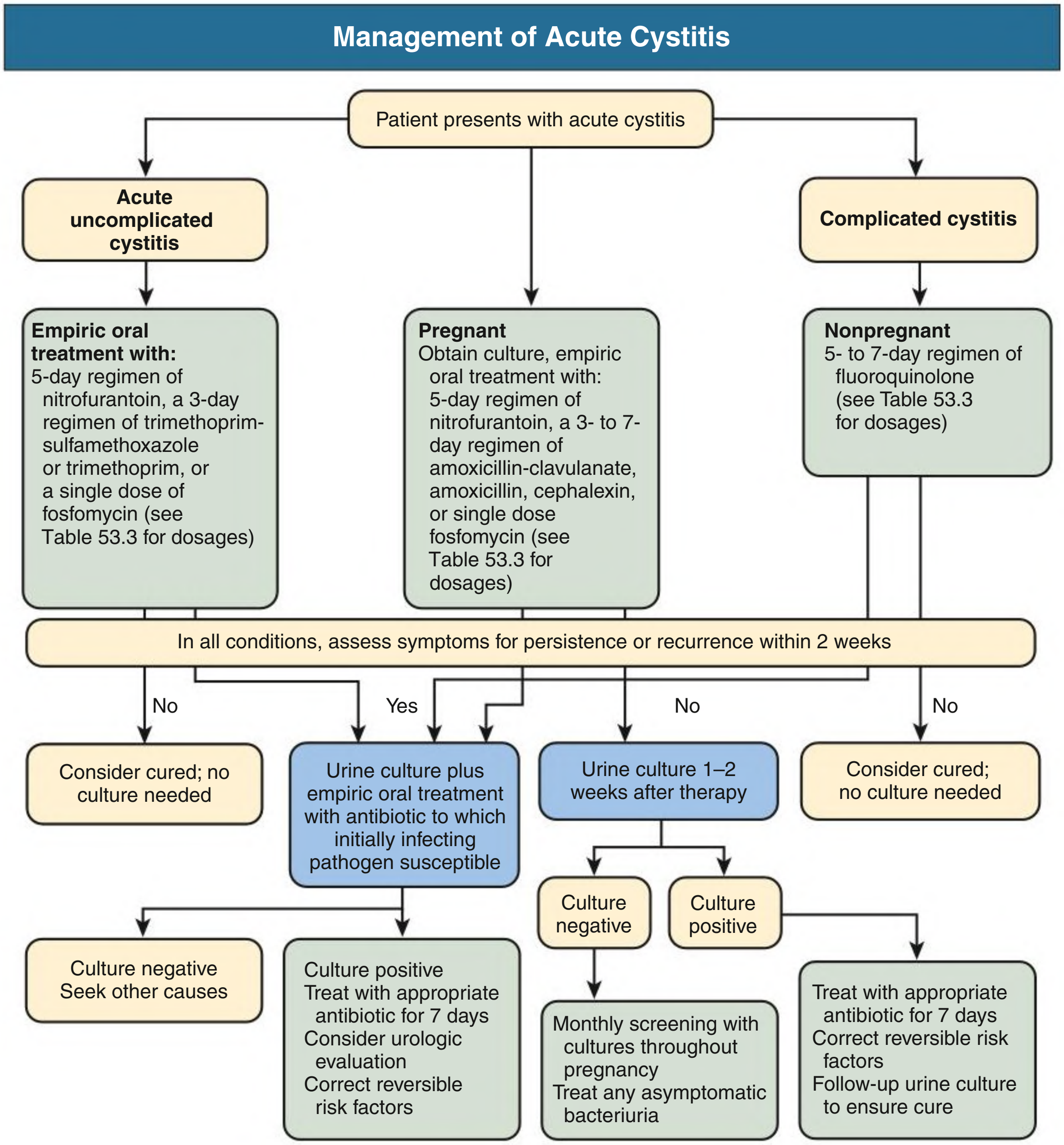

Fig. 53.1 Algorithm for management of acute cystitis.

TABLE 53.3 Oral Antimicrobial Agents for Acute Uncomplicated Cystitis

Drug	Dose (mg)	Interval[a]	Comment
Nitrofurantoin			Less active against *Proteus* spp.
Monohydrate/macrocrystals	100	q12h	
Macrocrystals	50	q6h	
TMP-SMX	160/800	q12h	If used in pregnancy (not approved use), avoid in first trimester.
Trimethoprim	100	q12h	If used in pregnancy (not approved use), avoid in first trimester.
Fosfomycin	3000	Single dose	Less effective than fluoroquinolone or TMP-SMX.
Pivmecillinam	400	q8–12h	Availability limited to some European countries; not available for use in North America. Associated with minimal resistance and propensity for collateral damage, but efficacy rates are lower than other agents.
Cefpodoxime proxetil	100	q12h	Comparable to TMP-SMX, inferior to ciprofloxacin in 3-day regimen.[28]
Amoxicillin-clavulanate	500/125	q12h	Inferior to ciprofloxacin in 3-day regimen.[29]
Amoxicillin	500	q12h	Used only when causative pathogen is known to be susceptible or for empiric treatment of mild cystitis in pregnancy.
Fluoroquinolones			
Ciprofloxacin	250	q12h	Avoid fluoroquinolones if possible in pregnancy, nursing mothers, or persons younger than 18 years old. Although highly effective, should be considered second-line treatment to preserve their usefulness for other infections. Moreover, in the United States, the FDA has stated that the risks of systemic fluoroquinolone antibacterial drugs outweigh their benefits for uncomplicated cystitis.
Ciprofloxacin extended release	500	q24h	
Levofloxacin	250	q24h	
Ofloxacin	200	q12h	

[a]Duration of therapy depends on the clinical setting (see text and Fig. 53.1).
FDA, U.S. Food and Drug Administration; TMP-SMX, trimethoprim-sulfamethoxazole.

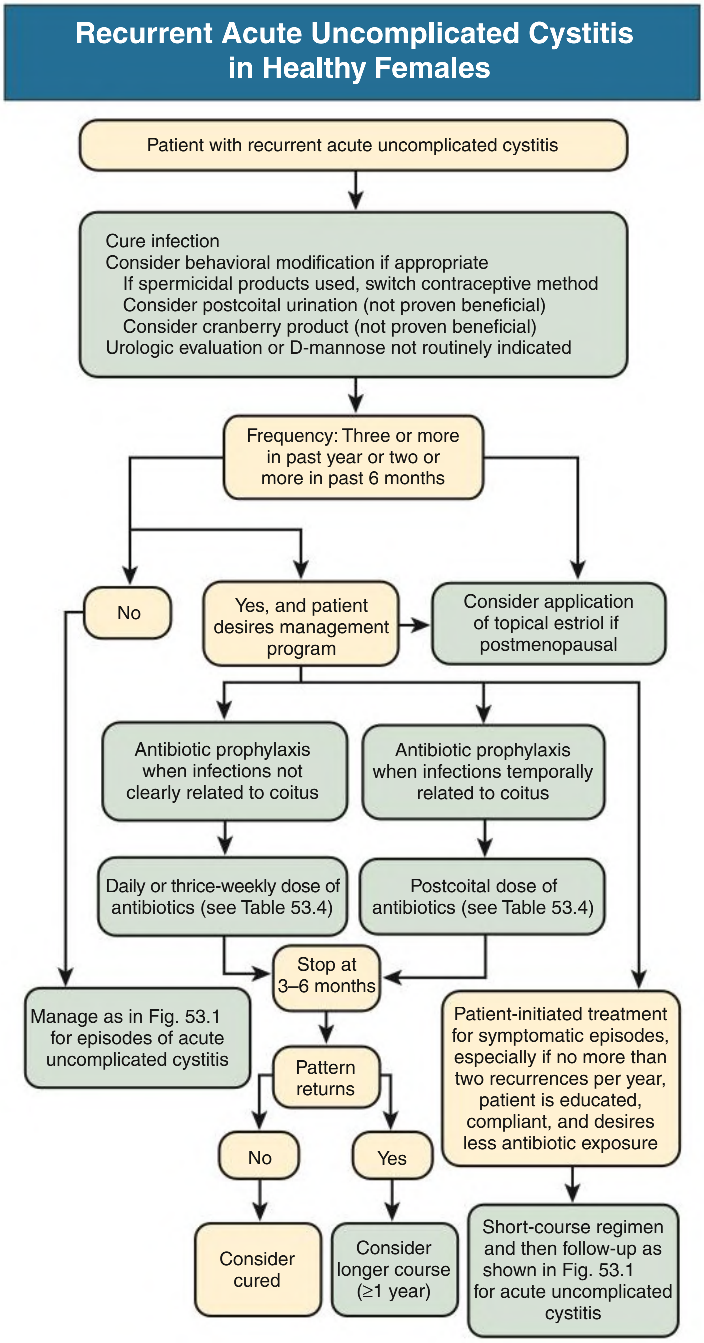

Fig. 53.2 Management strategies for recurrent acute uncomplicated cystitis.

Recurrent Acute Uncomplicated Cystitis in Females

Many cases of recurrent cystitis in healthy females are caused by persistence of the initially infecting strain in the fecal flora.[30] Studies in mice suggest that a latent reservoir of uropathogens in the bladder epithelium that persist after the initial UTI may be one cause of recurrence,[31] and indirect evidence indicates that this may occur in humans.[32] If the recurrence is within 1 or 2 weeks of treatment, an antimicrobial-resistant uropathogen should be considered, and a urine culture should be performed followed by treatment with an active antimicrobial. We treat later recurrences the same as the original infection, although if the recurrence is within 6 months, we consider a first-line drug other than the one used originally, especially if TMP-SMX was used, because of the likelihood of resistance.[33]

Long-term management of recurrent cystitis should be aimed to improve the quality of life while minimizing antimicrobial exposure.[13] Females with recurrent cystitis may benefit from behavioral modification (Fig. 53.2), such as avoiding spermicides, increasing fluid intake,[34] and ensuring postcoital micturition, although it is uncertain whether these practices are truly beneficial. Data on efficacy are sparse, but targeting the adherence mechanism of *Enterobacteriaceae* with D-mannose powder is another antimicrobial-sparing modality used by some females to prevent UTI.[35] Cranberry products are widely used to prevent recurrent UTI, but randomized, placebo-controlled trials (RCTs) have shown minimal benefits.[36] Females who do not want to try or who obtain no benefit from the preceding approaches should be offered

TABLE 53.4 Antimicrobial Prophylaxis Regimens for Females With Recurrent Acute Uncomplicated Cystitis

Drug	Dose (mg)	Frequency
Continuous Prophylaxis		
Nitrofurantoin	50 or 100	Daily
TMP-SMX	40/200	Daily
TMP-SMX	40/200	Three times weekly
Trimethoprim	100	Daily
Cefaclor	250	Daily
Cefalexin (cephalexin)	125 or 250	Daily
Postcoital Prophylaxis		
Nitrofurantoin	50 or 100	Single dose
TMP-SMX	40/200	Single dose
TMP-SMX	80/400	Single dose
Cephalexin	250	Single dose

See text and Fig. 53.2 for management strategy.
TMP-SMX, Trimethoprim-sulfamethoxazole.

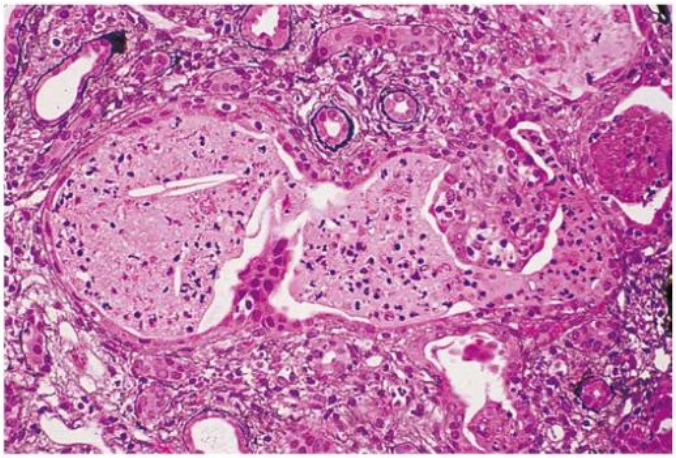

Fig. 53.3 **Acute Pyelonephritis.** Kidney tissue shows a dilated tubule with neutrophils enmeshed in proteinaceous debris ("pus casts") with adjacent interstitial inflammation. (Courtesy C. Alpers, University of Washington, Seattle.)

Fig. 53.4 **Acute Pyelonephritis.** Contrast-enhanced computed tomography scan shows areas of lower density caused by infection and edema *(arrows)*. (Courtesy W. Bush, University of Washington, Seattle.)

antimicrobial prophylaxis or have a course of antimicrobials on hand for intermittent self-treatment.

Antimicrobial prophylaxis reduces the risk for recurrent cystitis by approximately 95% (Table 53.4; see also Fig. 53.2).[37] Prophylaxis should be considered for females who experience three or more infections during a 12-month period or whenever the patient thinks their life is being adversely affected by frequent recurrences. Several approaches have been used, including continuous prophylaxis, postcoital prophylaxis (best for females who experience UTI only after intercourse), and intermittent self-treatment (which is really an early treatment method).[13] In postmenopausal females with recurrent UTI, intravaginal estradiol is effective, presumably by normalizing the vaginal flora, which reduces the risk for coliform colonization of the vagina.[38] This approach offers an alternative to antimicrobial strategies (see Fig. 53.2).

Promising antimicrobial-sparing approaches are being developed to specifically target virulence pathways, which might prevent uropathogens from causing disease, without altering the gut commensal microbiota. Antivirulence therapeutics target processes that are critical for UTI pathogenesis but are not required for the essential processes of growth and cell division (targets of conventional antimicrobials).

Acute Uncomplicated Pyelonephritis in Females and Males

Acute pyelonephritis is suggested by fever (temperature ≥ 38°C), chills, flank pain, nausea and vomiting, and costovertebral angle tenderness. Cystitis symptoms are variably present. Symptoms may vary from a mild illness to a sepsis syndrome with or without shock and kidney failure. Pyuria is almost always present, but leukocyte casts, specific for pyelonephritis, are infrequently seen. Gram stain of the urine sediment may aid in differentiating gram-positive and gram-negative infections, which can influence empiric therapy. A urine culture, which should be performed in all females with acute pyelonephritis, will have at least 10^4 cfu/mL of uropathogens in nearly all patients.[39]

Kidney biopsy shows focal inflammation with neutrophil and monocyte infiltrates, tubular damage, and interstitial edema (Fig. 53.3). On imaging, the infected kidney is often enlarged, and contrast-enhanced computed tomography (CT) shows decreased opacification of the affected parenchyma, typically in patchy, wedge-shaped, or linear patterns (Fig. 53.4).

The availability of effective oral antimicrobials, especially fluoroquinolones, allows initial oral therapy in appropriate patients or, in those initially requiring parenteral therapy, the timely conversion from intravenous to oral therapy. Indications for hospital admission include inability to maintain oral hydration or to take medications; uncertain social situation or concern about compliance; uncertainty about the diagnosis; and severe illness with high fevers, severe pain, and marked debility. Outpatient therapy is safe and effective for select patients who can be stabilized with parenteral fluids and antibiotics in an urgent care facility and sent home with oral antibiotics under close supervision. In one population-based study of acute pyelonephritis in adult females, only 7% were hospitalized.[4]

The management strategy for acute uncomplicated pyelonephritis is shown in Fig. 53.5. Many effective parenteral (Table 53.5) and oral

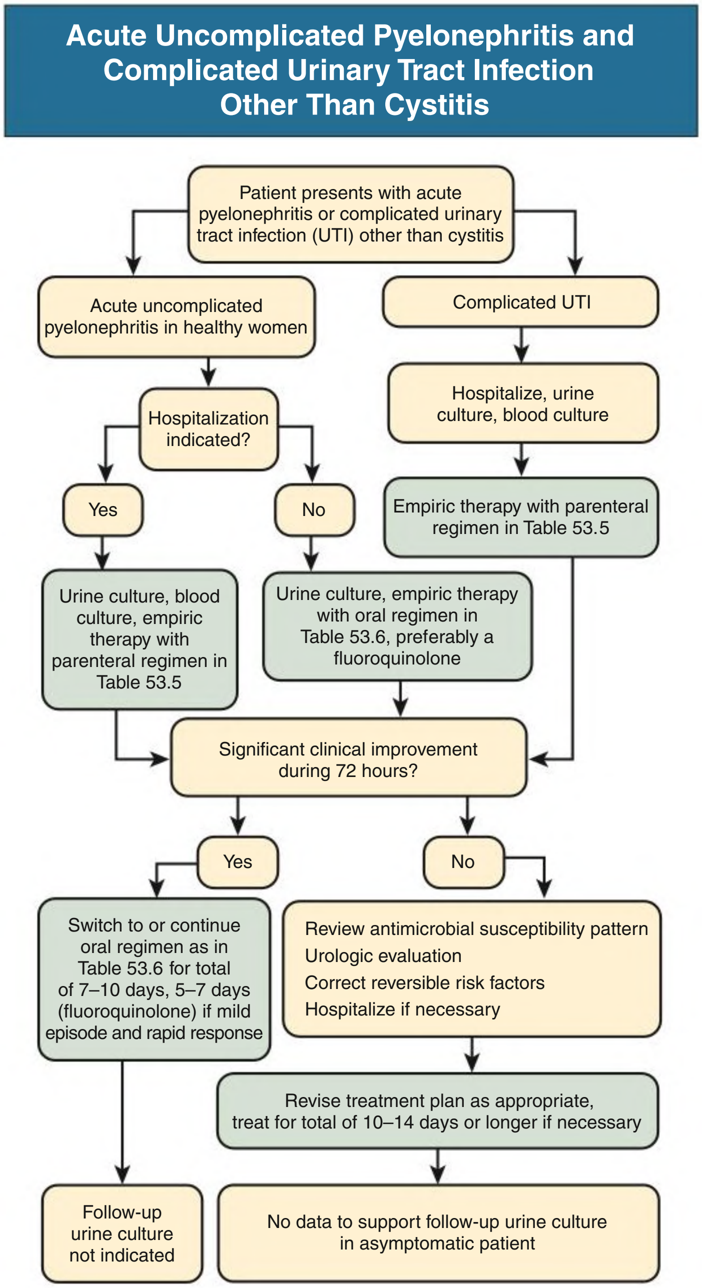

Fig. 53.5 Management algorithm for acute uncomplicated pyelonephritis and complicated urinary tract infection other than cystitis.

(Table 53.6) regimens are available for patients with acute uncomplicated pyelonephritis. For outpatients, an oral fluoroquinolone should be used for initial empiric treatment of infection caused by gram-negative bacilli.[21] TMP-SMX or other agents can be used if the infecting strain is known to be susceptible. If enterococci are suspected from the Gram stain, amoxicillin should be added to the treatment regimen until the causative organism is identified. Second- and third-generation cephalosporins also appear effective, although published data are sparse. Nitrofurantoin, fosfomycin, and pivmecillinam are not approved or recommended for the treatment of pyelonephritis because they only achieve therapeutic levels in the bladder. When antimicrobial resistance or intolerance of oral medications is a concern, one or more doses of a broad-spectrum parenteral antimicrobial is recommended until in vitro activity can be ensured.[21,39]

For hospitalized patients without evidence of gram-positive infection, ceftriaxone is effective and inexpensive. If enterococci are suspected based on the Gram stain, ampicillin plus gentamicin, ampicillin-sulbactam, or piperacillin-tazobactam are reasonable empiric choices. TMP-SMX should not be used as empiric monotherapy for pyelonephritis in areas with a high prevalence of resistance to this agent. Patients with acute uncomplicated pyelonephritis often can be switched to oral therapy after 24 to 48 hours, although longer durations of parenteral therapy are occasionally indicated in patients with continued high fever, severe flank pain, or persistent nausea and vomiting.

Treatment of acute uncomplicated pyelonephritis can be limited to 5 to 7 days for patients who have a rapid resolution of fever and

TABLE 53.5 Parenteral Regimens for Acute Uncomplicated Pyelonephritis and Complicated Urinary Tract Infection[a]

Drug	Dose (mg)	Interval
Ceftriaxone	1000–2000	q24h
Cefepime	1000–2000	q12h
Fluoroquinolones[b]		
Ciprofloxacin	200–400	q12h
Levofloxacin	250–750	q24h
Gentamicin[b] (± ampicillin)	3–5 mg/kg body weight	q24h
	1 mg/kg body weight	q8h
Ampicillin (+ gentamicin[b])	1000	q6h
Trimethoprim-sulfamethoxazole[b]	160/800	q12h
Aztreonam	1000	q8–12h
Piperacillin-tazobactam	3375	q6–8h
Imipenem-cilastatin[b,c]	250–500	q6–8h
Meropenem[c]	500	q8h
Ertapenem[c]	1000	q24h
Ceftolozane/tazobactam	1500	q8h
Ceftazidime/avibactam	2500	q8h
Vancomycin[d]	1000	q12h

[a]Duration depends on clinical setting (see text and Fig. 53.5).
[b]Avoid, if possible, in pregnancy.
[c]Recommended if ESBL Enterobacteriacea is suspected or known. Ertapenem is not indicated for suspected or known *Pseudomonas* infection.
[d]Recommended if methicillin-resistant *Staphylococcus aureus* is suspected or known.

TABLE 53.6 Oral Regimens for Acute Uncomplicated Pyelonephritis and Complicated Urinary Tract Infection[a,b]

Drug	Dose (mg)	Interval	Comment
Fluoroquinolones			Preferred for empiric treatment; avoid if possible in pregnancy, nursing mothers, or persons younger than 18 years of age.
Ciprofloxacin	500	q12h	
Ciprofloxacin extended release	1000	q24h	
Levofloxacin	250–750	q24h	
Trimethoprim-sulfamethoxazole	160/800	q12h	Use only when the causative pathogen is known to be susceptible. If used in pregnancy (not approved use), avoid in first trimester.
Cefpodoxime proxetil	200	q12h	Data are sparse; use only when the causative pathogen is known to be susceptible.
Amoxicillin-clavulanate	500/125–875/125	q12h	Use only when the causative pathogen is known to be susceptible or in addition to a broad-spectrum agent when empiric coverage against enterococci is desirable.

[a]A long-acting parenteral antibiotic should be given concomitantly if there are concerns about drug resistance.
[b]Duration depends on clinical setting (see text and Fig. 53.5).

symptoms soon after initiation of treatment. However, β-lactam regimens shorter than 14 days have been associated with unacceptably high failure rates.[40] One study demonstrated superiority of a 7-day ciprofloxacin regimen over a 14-day TMP-SMX regimen, with the difference accounted for entirely by the higher rate of resistance of the uropathogens to TMP-SMX.[33]

A U.S. study of patients with acute pyelonephritis presenting to emergency departments demonstrated that the prevalence of fluoroquinolone-resistant and ESBL-producing strains is increasing and that such patients are often treated with empiric antimicrobial drugs that are inactive against the causative strains.[41] The authors concluded that in areas with high fluoroquinolone resistance rates, where ESBL-producing *Enterobacteriaceae* infections have emerged, or among persons with antimicrobial drug resistance risk factors, clinicians should consider empiric treatment with a carbapenem or another agent with activity against these resistant strains.

Routine posttreatment urine cultures in asymptomatic patients are not useful, but cultures should be performed if symptoms persist or recur. Recurrent infections are treated with a 7- to 14-day course of an antibiotic to which the organism is susceptible. Symptomatic patients who have persistent infection with the same strain as the initial infecting strain warrant at least 10 to 14 days of therapy with a different antibiotic, and complicating factors should be sought and corrected if possible.

Males With Cystitis

Among premenopausal females, cystitis is more common than among similarly aged males. However, after the age of 50 to 60, rates of cystitis in males increase, although never reaching the rates of females of equivalent age.[42] Whereas cystitis in females is well studied, similar studies of male patients are lacking. Males with cystitis are often considered to be a category of complicated UTI, with male sex considered a factor that increase the risk of treatment failure or recurrence. Whether this classification is applicable to all males is unproven; many seem to respond to traditional cystitis treatment but may require a longer duration of treatment than females. Observational studies[43,44] and a recent trial[44a] suggest that shorter-duration therapy performs similarly well to longer-duration therapy, although even the shorter-duration therapy in these studies was longer than the 1- to 5-day regimens that are effective for females with cystitis. Whether males with cystitis who have additional complicating factors (e.g., urinary retention, renal calculi, indwelling catheters) need longer-duration therapy is unknown.

Another difference between males and females with cystitis is the relative lack of predictability regarding the causative organisms. Whereas the microbiology of cystitis among females is predictable, among males there is more variability. *E. coli* is still the most commonly isolated pathogen, but in some series is only a plurality versus the clear majority observed in females.[45,46] Because of this unpredictability, a urine culture should always be obtained to help guide therapy in male patients with cystitis.

Complicated Infections

Patients with UTI in the presence of complicating conditions may present with classic signs of cystitis and pyelonephritis but also may have vague or nonspecific symptoms, such as fatigue, irritability, nausea, headache, and abdominal or back pain. Acute cystitis in healthy individuals other than young females is more likely to involve occult renal or prostatic infection and may respond poorly to short-course therapy. Noninvasive tools to localize infections to the kidney or prostate are lacking, so clinical estimation of risk in a given patient is imprecise. Some patients, such as those who are diabetic or pregnant, warrant special attention because of the serious complications that can occur if treatment is inadequate. Urethritis should be excluded in dysuric sexually active males by a urethral Gram stain or a urine nucleic acid amplification test for *Neisseria gonorrhoeae* and *Chlamydia trachomatis*. Complicated UTI, just as with uncomplicated infection, is generally associated with pyuria and bacteriuria, although these may be absent if the infection does not communicate with the collecting system.

Urine culture should always be performed in patients with suspected complicated UTI. Traditionally complicated UTI is at least 10^5 cfu/mL in the urine of females and at least 10^4 cfu/mL in males, but lower counts in symptomatic persons, as demonstrated in patients with uncomplicated UTI, may well represent significant bacteriuria. This is especially true when the specimen is collected from a urinary catheter. Thus, it is reasonable to use a colony count threshold of greater than 10^3 cfu/mL of uropathogens to diagnose complicated UTI.

The wide variety of underlying conditions (see Box 53.1), diverse bacterial agents (see Table 53.2), and paucity of RCTs make generalizations about antimicrobial therapy difficult. Figs. 53.1 and 53.5 outline the management strategy for complicated cystitis and other complicated UTIs, respectively.

Correction of any underlying anatomic, functional, or metabolic defect should be attempted because antibiotics alone may not be successful.[1] For empiric therapy in patients with mild to moderate illness who can be treated with oral medication, the fluoroquinolones provide the broadest spectrum of antimicrobial activity, cover most expected pathogens, and achieve high levels in the urine and urinary tract tissue. An exception is moxifloxacin, which may not achieve sufficient concentrations in urine to be effective for complicated UTI. If the infecting pathogen is known to be susceptible, TMP-SMX or other agents are reasonable therapeutic choices. Nitrofurantoin and fosfomycin should be avoided except for cystitis in pregnancy, in which duration of treatment is 5 days or single-dose, respectively.

For initial treatment in more seriously ill, hospitalized patients, several parenteral antimicrobial agents are available (see Table 53.5). The broader spectrum agents shown in Table 53.5 are preferable for health care–associated infections. *S. aureus* is more common in complicated UTIs, and, if suspected, the therapeutic regimen should have activity against *S. aureus*. Studies show that a high proportion of *S. aureus* isolates, even in the community, are methicillin resistant (MRSA), so vancomycin should be included in the empiric treatment regimen if *S. aureus* is suspected. Factors that must be considered in the management of complicated UTI include the increasing prevalence of resistance to fluoroquinolones in institutional settings and the frequency of enterococcal infections.

The antimicrobial regimen can be modified when the infecting strain has been identified and antimicrobial susceptibilities are known. Patients receiving parenteral therapy can be switched to oral treatment after clinical improvement. Few studies evaluate duration of treatment in populations with complicated UTIs. However, it is desirable to limit the duration of treatment, especially for milder infections, to reduce the selection pressure for drug-resistant flora. In one study, clinical and microbiologic success rates after treatment were almost identical in patients with acute pyelonephritis or complicated UTI treated with a 5-day course of levofloxacin or a 10-day course of ciprofloxacin.[47] These data suggest that a 5- to 10-day regimen is reasonable for most patients with complicated UTI, depending on their severity of illness and clinical response; shorter regimens, such as a 5-day regimen of a urinary fluoroquinolone, are likely to be sufficient in patients who are less severely ill, are infected with uropathogens susceptible to the antibiotic used, and have a rapid response to treatment. A recent large retrospective study of male veterans with UTI found no difference in recurrence rates with 7 days of treatment versus longer, with a trend toward more *Clostridium difficile* infections in those treated longer.[43] A clinical trial of V in veterans with UTI but without fever confirmed non-inferiority of 7 versus 14 days of antibiotics.[44a] At least 10 to 14 days of therapy is recommended in patients who have a delayed response.

Routine posttreatment cultures are not indicated unless the patient is symptomatic, except in pregnant patients (see Fig. 53.1). In males, early recurrence of UTI with the same species suggests a prostatic source of infection and warrants a 4- to 6-week regimen of either a fluoroquinolone (preferable) or TMP-SMX, depending on the antimicrobial susceptibility of the infecting strain.

Chronic Kidney Disease

Because of the wide variety of underlying diseases and comorbidities, prior instrumentation of the urinary tract, and differences in age and sex, the incidence of UTI in patients with CKD is not known.[48] Moreover, there are few data on urine concentrations of antimicrobials used for the treatment of UTI in patients with CKD.[49] Studies in animals suggest that (1) urine drug concentrations are necessary to sterilize urine, (2) effective tissue concentrations are necessary to treat pyelonephritis, and (3) serum concentrations of antimicrobials are correlated with the drug concentrations in kidney tissue. Thus, for patients with CKD who have a therapeutic serum drug level and adequate perfusion of the kidney parenchyma, the delivery of therapeutic concentrations of both the parenchyma and the urine should be adequate, but some oral agents for cystitis may not deliver adequate concentrations to the urine.[49] For patients with a low glomerular filtration rate (GFR) and pyelonephritis or cystitis, the agents listed in Tables 53.5 and 53.6 for pyelonephritis and complicated cystitis should be adequate to treat these infections if the organism is susceptible.[49]

As noted previously, β-lactams are not as effective as fluoroquinolones, even in patients with normal kidney function.[28,29] Doses of ciprofloxacin or levofloxacin should be adjusted for GFR; moxifloxacin is not recommended for UTI treatment. Nitrofurantoin and sulfamethoxazole are not recommended in patients with reduced GFR, although trimethoprim concentrations appear to be adequate.[48] Likewise reduced kidney function (such as CKD stage 3 or higher) significantly decreases the excretion of fosfomycin, which also should not be used in such patients.

Catheter-Associated Infections

Approximately 15% to 25% of patients in general hospitals have a urethral catheter inserted at some time during their stay, and approximately 5% to 10% of long-term care facility residents are managed with urethral catheterization, in some cases for years. The incidence of bacteriuria associated with indwelling catheters is 3% to 10% per day of catheterization, and the duration of catheterization is the most important risk factor for the development of catheter-associated bacteriuria.

Catheter-associated bacteriuria is the most common source of gram-negative bacteremia in hospitalized patients, often associated with catheter obstruction or trauma. Complications of long-term catheterization (≥30 days) include almost universal bacteriuria, often with multiple antibiotic-resistant flora, and (in addition to cystitis, pyelonephritis, and bacteremia, as seen with short-term catheterization) frequent febrile episodes, catheter obstruction, stone formation associated with urease-producing uropathogens, and local genitourinary infections. Other rare complications include fistula formation and bladder cancer. An increase in mortality risk has been reported with catheter-associated bacteriuria, but it is difficult to distinguish the role of the catheter because most deaths occur in patients who have severe underlying disease.

Most episodes of catheter-associated bacteriuria are asymptomatic and do not require routine screening or treatment because treatment does not reduce the complications of bacteriuria and can lead to antimicrobial resistance.[7] Moreover, the presence or absence of pyuria does not differentiate symptomatic from asymptomatic urinary infection.[7] Symptomatic UTIs, often caused by many multidrug-resistant uropathogens, warrant broad-spectrum therapy, as described previously. In a symptomatic catheterized patient, a urine culture specimen should be obtained from a freshly placed catheter if the catheter has been in place for a few days because the catheter biofilm may result in spurious culture results. Moreover, clinical outcomes are improved if the catheter is replaced at the time of antimicrobial therapy.[50] The recommended duration of antimicrobial treatment for patients who have prompt resolution of symptoms is 7 days, and 10 to 14 days if response is delayed.[51]

Preventive measures are indicated to reduce the morbidity, mortality, and costs of catheter-associated infection. Effective strategies include avoidance of a catheter when possible and, when the catheter is necessary, sterile insertion, prompt removal, and strict adherence to a closed collecting system.[6,51] Bundling of preventive steps may be useful.[52,53] Meta-analyses have shown that rates of catheter-associated bacteriuria, at least in patients catheterized for less than 2 weeks, are higher in patients with an indwelling urethral catheter than in those with intermittent or suprapubic catheterization.[54] Likewise, condom catheterization is preferable to indwelling urethral catheterization in appropriately selected males.[54] Although highly effective in reducing catheter-associated bacteriuria rates, prophylactic systemic antimicrobial agents are generally not recommended for routine use because of concerns about selection for antimicrobial resistance. Antimicrobial-coated catheters appear to be effective in reducing catheter-associated bacteriuria in patients catheterized for less than 2 weeks, but these have not been shown to be effective in reducing symptomatic infection.

Spinal Cord Injury

Spinal cord injury alters the dynamics of voiding and often requires the use of bladder drainage with catheters. The diagnosis of UTI in patients with spinal cord injuries is based on the combination of symptoms and signs (which are often nonspecific), pyuria, and significant bacteriuria. Uropathogens are usually present in quantities of at least 10^5 cfu/mL. Fluoroquinolones are the empiric oral agents of choice in patients with mild to moderate infection, although many uropathogens, even in the outpatient setting, are resistant to this class of antibiotic, and parenteral antibiotics may be needed.

Treatment of asymptomatic bacteriuria in patients with spinal cord injuries is not of proven benefit and increases the risk for infection with antimicrobial-resistant uropathogens.[7,55] Likewise, antibiotic prophylaxis is generally not recommended, although it may be considered for select outpatients with frequent symptomatic UTIs for whom there are no correctible risk factors. Recent studies have shown that use of a hydrophilic-coated catheter for intermittent catheterization is associated with a reduction in the incidence of symptomatic UTI in patients with spinal cord injury.[56]

Prostatitis

Prostatitis occurs in up to 25% of males during their lifetime, but it is caused by acute or chronic bacterial infection in a minority.[57] The most common organisms causing bacterial prostatitis are gram-negative bacilli, including *E. coli, Proteus* spp., *Klebsiella* spp., *P. aeruginosa,* and, less frequently, enterococci and *S. aureus.* The pathogenesis of prostatitis is believed to be related to reflux of infected urine from the urethra into the prostatic ducts. Prostatic calculi, commonly found in adult males, may provide a nidus for bacteria and protection from antibacterial agents.

Acute bacterial prostatitis is rare. Patients present with dysuria, frequency, urgency, obstructive voiding symptoms, fever, chills, and myalgias. The prostate is tender and swollen, and the prostate exam should be limited to assessment of size and tenderness to avoid precipitating bacteremia. The patient will usually have pyuria and a positive urine culture. Patients who are severely ill require hospitalization and parenteral antibiotics, but many patients can be treated in the outpatient setting with oral fluoroquinolones. The recommended duration of treatment is 14 to 30 days.[57] Rarely, abscess formation may occur, which should be treated by source control with the assistance of a urologist.

Chronic bacterial prostatitis is characterized by recurrent UTIs with the same uropathogen with intervening asymptomatic periods. The prostate typically is normal to palpation during asymptomatic periods. Chronic bacterial prostatitis is characterized microscopically by the presence of at least 10 leukocytes per high-power field in expressed prostatic secretions or postmassage voided urine in the absence of significant pyuria in first-voided and midstream urine specimens, as well as a uropathogen colony count at least 10-fold higher in the expressed prostatic secretions or postmassage voided urine compared with the first-voided midstream urine. In addition, macrophage-laden fat droplets (oval fat bodies) are usually prominent in the prostatic secretions. These tests, however, are infrequently performed by urologists. Cure rates, which historically have been low, are 60% to 80% with the fluoroquinolones, which are the antibiotics of choice. The optimal duration of treatment is unknown, but 4 to 6 weeks is recommended by some authorities,[57] whereas others recommend up to 3 months. Some patients require long-term, low-dose suppressive therapy to prevent symptomatic UTIs. Surgical intervention is only rarely considered and is associated with high morbidity.

Kidney Abscess

Kidney cortical and corticomedullary abscesses and perirenal abscesses occur in 1 to 10 per 10,000 hospital admissions.[58] Patients usually present with fever, chills, back or abdominal pain, and costovertebral angle tenderness, but they may have no urinary symptoms or findings if the abscess does not communicate with the collecting system, as often occurs with a cortical abscess. Bacteremia may be primary (cortical abscess) or secondary (corticomedullary or perirenal). The clinical presentation may be insidious and nonspecific, especially with perirenal abscess, and the diagnosis may not be made until admission to a hospital or at autopsy. CT is recommended to establish the diagnosis and location of a renal or perirenal abscess (Fig. 53.6). Empiric antibiotic therapy should be broad and cover *S. aureus* and other uropathogens causing complicated UTI (see Fig. 53.5 and Table 53.5) and modified once urine culture results are known.

A *renal cortical abscess* (kidney carbuncle) is usually caused by *S. aureus,* which reaches the kidney by hematogenous spread. Treatment with antibiotics is usually effective, and drainage is not required unless the patient is slow to respond. A *renal corticomedullary abscess,* in contrast, usually results from ascending UTI in association with an underlying urinary tract abnormality, such as obstructive uropathy or VUR, and is usually caused by common uropathogenic species such as *E. coli* and other gram-negative bacilli. Such abscesses may extend deep into the renal parenchyma, perforate the renal capsule, and form

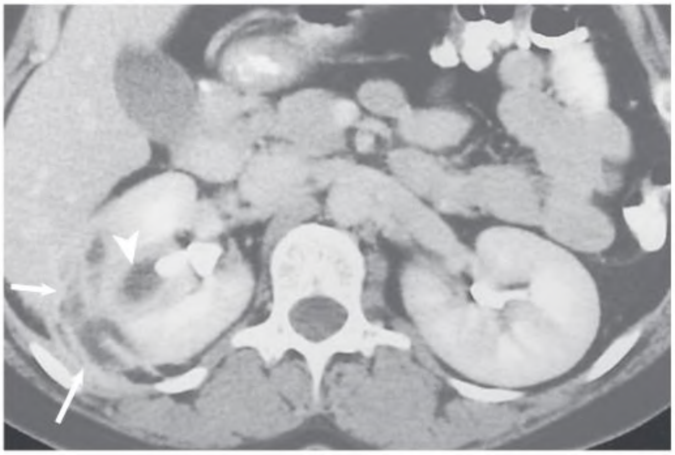

Fig. 53.6 **Renal Abscess.** Contrast-enhanced computed tomography scan shows an abscess in the medulla of the kidney *(arrowhead)* with penetration and extension into the perinephric space *(arrows)*. (Courtesy L. Towner.)

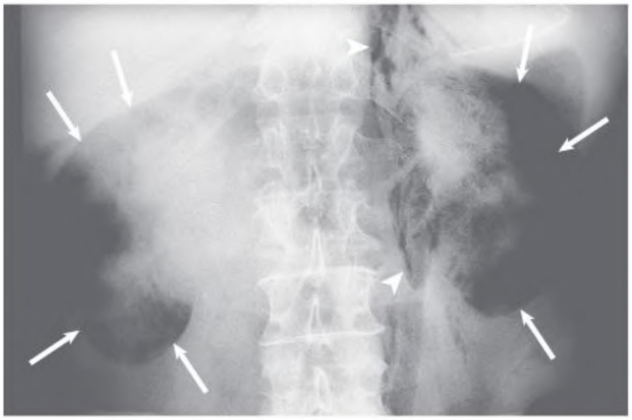

Fig. 53.7 **Emphysematous Pyelonephritis.** A plain radiograph in this febrile patient with diabetes reveals diffuse gas formation throughout both kidneys *(outlined by arrows)* and gas dissecting in the left retroperitoneal space *(arrowheads)*. (Courtesy W. Bush, University of Washington, Seattle.)

a perirenal abscess. Treatment with antimicrobial agents without drainage may be effective if the abscess is small and if the underlying urinary tract abnormality can be corrected. Aspiration of the abscess may be necessary in some patients, and nephrectomy may occasionally be required in patients with diffuse renal involvement or with severe sepsis. Perirenal abscesses usually occur in the setting of obstruction or other complicating factors (see Box 53.1) and result from ruptured intrarenal abscesses, hematogenous spread, or spread from a contiguous infection. Causative uropathogens are those usually found in complicated UTIs (see Table 53.2), including *S. aureus* and enterococci; polymicrobial infections are common. Anaerobes or *Mycobacterium tuberculosis* may be causative (see Chapter 54). A previously high mortality rate has been lowered with earlier diagnosis and therapy. In contrast to the other types of renal abscesses, drainage of pus is the cornerstone of therapy and nephrectomy may be indicated.

Papillary Necrosis

More than half of patients who develop papillary necrosis have diabetes, almost always in conjunction with a UTI, but the condition also complicates sickle cell disease, analgesic abuse, and obstruction. Kidney papillae are vulnerable to ischemia because of the sluggish blood flow in the vasa recta, and relatively modest ischemic insults may cause papillary necrosis. The clinical features are those typical of pyelonephritis. In addition, passage of sloughed papillae into the ureter may cause renal colic, renal impairment or failure, or obstruction with severe urosepsis. Papillary necrosis in the setting of pyelonephritis is associated with pyuria and a positive urine culture. Causative uropathogens are those typical of complicated UTI. CT is the preferred diagnostic procedure. Radiologic findings include an irregular papillary tip; dilated calyceal fornix; extension of contrast material into the parenchyma; and a separated crescent-shaped papilla surrounded by contrast called the *ring sign* (see Chapter 51, Fig. 51.8). Broad-spectrum antibiotics are indicated. Papillae obstructing the ureter may require removal with a cystoscopic ureteral basket or relief of obstruction by insertion of a ureteral stent.

Emphysematous Pyelonephritis

Emphysematous pyelonephritis is a fulminant, necrotizing, life-threatening variant of acute pyelonephritis caused by gas-forming organisms, including *E. coli, Klebsiella pneumoniae, P. aeruginosa,* and *Proteus mirabilis.*[59] Up to 90% of cases occur in diabetic patients, and obstruction may be present. Symptoms are suggestive of pyelonephritis, and there may be a flank mass. Dehydration and ketoacidosis are common. Pyuria and a positive urine culture are usually present. Gas is usually detected by a plain abdominal radiograph or ultrasound (Fig. 53.7). CT is the diagnostic modality of choice, however, because it can localize the gas better than ultrasound. Accurate localization of gas is important because gas also may form in an infected obstructed collecting system or renal abscess; although serious, these conditions do not carry the same poor prognosis and are managed differently. Parenteral broad-spectrum antibiotics and percutaneous catheter drainage with relief of obstruction may be adequate for less severely ill patients, but nephrectomy is warranted for those who are more severely ill and those less severely ill who do not respond to the preceding steps. Medical treatment is associated with mortality of 60% to 80%, which is lowered to 20% or less with surgical intervention (e.g., nephrectomy, percutaneous drainage).

Kidney Malacoplakia

Malacoplakia is a chronic granulomatous disorder of unknown etiology involving the genitourinary, gastrointestinal, skin, and pulmonary systems.[60] It is characterized by an unusual inflammatory reaction to a variety of infections and is manifested by the accumulation of macrophages containing calcified bacterial debris called *Michaelis-Gutmann bodies* (Fig. 53.8). The underlying disorder appears to be a monocyte-macrophage bactericidal defect. The diagnosis is made by histologic examination of involved tissue. Genitourinary malacoplakia, most often involving the bladder, is usually associated with gram-negative UTI. Patients with kidney malacoplakia generally have fever, flank pain, pyuria and hematuria, bacteriuria, and, if both kidneys are involved, impaired renal function. CT usually shows enlarged kidneys with areas of poor enhancement, and the condition may be indistinguishable from other infectious or neoplastic lesions. On occasion, the malacoplakia may extend through the kidney capsule into the perinephric space, simulating a renal carcinoma (see Fig. 53.8). Treatment consists of therapy with a broad-spectrum antimicrobial, attempted correction of any underlying complicating conditions, and improvement of renal function. Nephrectomy is recommended for advanced unilateral disease. When the disease is bilateral or occurs in a transplanted kidney, the patient's prognosis is very poor.

Xanthogranulomatous Pyelonephritis

Xanthogranulomatous pyelonephritis is a poorly understood, uncommon but severe chronic destructive granulomatous inflammation of

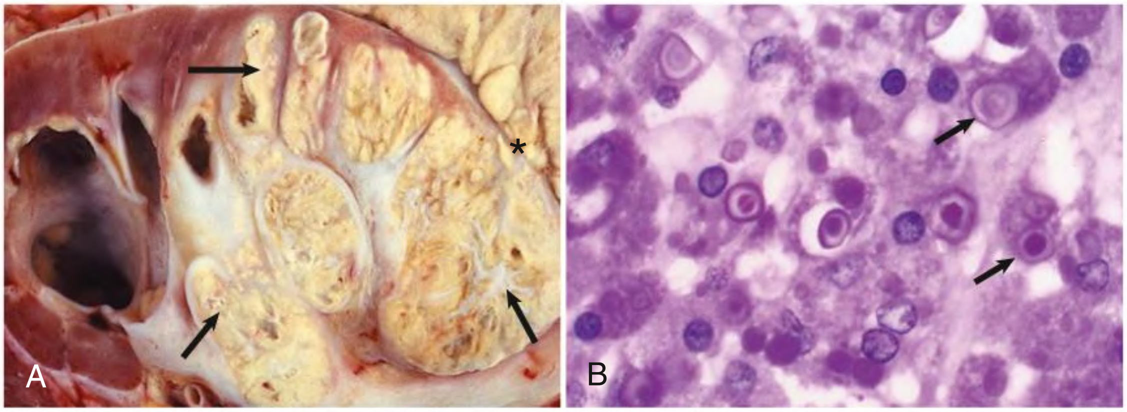

Fig. 53.8 **Renal Malacoplakia.** (A) Malacoplakia involving most of the kidney *(arrows)* with extension through the capsule *(asterisks)*. A small portion of normal kidney is present associated with hydronephrosis secondary to obstruction by the malacoplakia. (B) The kidney tissue shows many macrophages containing intracytoplasmic inclusions (*arrows* identify two particularly well-demarcated macrophages with Michaelis-Gutmann bodies). (Courtesy L. Truong, Baylor College of Medicine, Houston, and N. Sheerin, Guy's Hospital, London.)

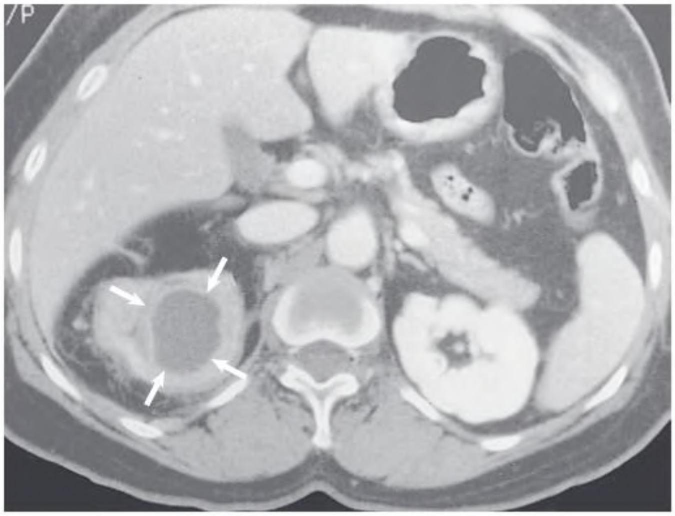

Fig. 53.9 **Xanthogranulomatous Pyelonephritis.** Contrast-enhanced computed tomography scan with an inflammatory mass outlined by *arrows.* Pathologic diagnosis confirmed xanthogranulomatous pyelonephritis. (Courtesy W. Bush, University of Washington, Seattle.)

the kidney parenchyma associated with obstruction and infection of the urinary tract.[61] The renal parenchyma is replaced with a diffuse or segmental cellular infiltrate of foam cells, which are lipid-laden macrophages. The process also may extend beyond the kidney capsule to the retroperitoneum. Its pathogenesis appears to be multifactorial, with infection complicating obstruction and leading to ischemia, tissue destruction, and accumulation of lipid deposits. Patients with xanthogranulomatous pyelonephritis are characteristically middle-aged females and have chronic symptoms such as flank pain, fever, chills, and malaise. Flank tenderness, a palpable mass, and irritative voiding symptoms are common. The urine culture is usually positive with *E. coli,* other gram-negative bacilli, or *S. aureus.* CT generally shows an enlarged nonfunctioning kidney, often the presence of calculi and low-density masses (xanthomatous tissue), and, in some cases, involvement of adjacent structures (Fig. 53.9). It may be difficult to distinguish from neoplastic disease. Broad-spectrum antimicrobials are indicated, but total or partial nephrectomy is usually necessary for cure.

Asymptomatic Bacteriuria

Asymptomatic bacteriuria is common and generally benign.[7] Pyuria is almost universally present, especially in elderly people, and is a predictor for subsequent symptomatic UTI in some groups. Causative uropathogens are the same as those causing UTIs in the same population. Testing for and treatment of asymptomatic bacteriuria is generally not warranted.[7] In young females with recurrent UTI, asymptomatic bacteriuria may paradoxically be protective against symptomatic recurrence and treatment may increase the risk for such recurrences.[62] Two exceptions to this are pregnant patients and patients undergoing urologic surgery; in both these groups, screening for and treatment of asymptomatic bacteriuria is recommended to prevent subsequent complications.[7] Current management strategies in patients with a renal transplant, including long-term antimicrobial prophylaxis, help prevent both asymptomatic bacteriuria and symptomatic UTI. It is not clear, however, whether testing for or treatment of asymptomatic bacteriuria in such patients is worthwhile.[7] Some authorities advise treatment of asymptomatic bacteriuria found in patients with anatomic or functional abnormalities of the urinary tract, diabetic patients, and patients with urea-splitting bacteria (e.g., *P. mirabilis, Klebsiella* spp.). Evidence-based guidelines for evaluation and treatment of asymptomatic bacteriuria in these populations are needed.

Asymptomatic bacteriuria in catheterized patients in hospitals and long-term care facilities, although thought to be generally benign, represents a large reservoir of antimicrobial-resistant urinary pathogens that increases the risk for cross-infection among catheterized patients and results in frequent inappropriate antimicrobial use.[51]

IMAGING OF THE URINARY TRACT

Urologic consultation and evaluation of the urinary tract should be considered in patients who present with symptoms or signs of obstruction, urolithiasis, flank mass, or urosepsis. Similarly, such an evaluation should be considered for those patients with presumptive uncomplicated or complicated UTI who have *not* had a satisfactory clinical response after 72 hours of treatment, to exclude complicating factors. A kidney ultrasound can detect the size and contour of the kidneys and bladder, the presence of a renal mass or abscess, certain renal and ureteral calculi, hydronephrosis suggestive of obstructive uropathy, and elevated postvoid residual urine.[63] A plain abdominal radiograph (kidneys, ureters, bladder [KUB]) can identify radiopaque calculi along the genitourinary tract, especially proximal and distal ureteral stones that can be missed on ultrasound. Kidney ultrasound and KUB are less sensitive than CT for detection of many conditions in patients with complicated UTI. Any finding suggesting mass or complex fluid collection should prompt follow-up imaging with CT. CT offers fine anatomic detail and is thus the superior study for evaluation of focal inflammation, renal or perirenal abscess and masses, and both radiopaque and radiolucent stones. However, CT also carries the greatest risk profile, exposing the patient to both intravenous contrast and ionizing radiation.[63] Non–contrast-enhanced spiral CT is a rapid, safe, and sensitive method for evaluating patients with suspected kidney stones. Radionuclide imaging procedures have no role in the evaluation of adults with UTI, although they are useful in children with pyelonephritis (see Chapter 62).

Excretory urography and cystoscopy in females with recurrent cystitis rarely demonstrate abnormalities or alter management[3] and therefore are not recommended. Likewise, imaging studies in young females with acute pyelonephritis are also generally not cost-effective and have a low diagnostic yield, although it is reasonable to obtain such studies after two episodes of pyelonephritis or if any complicating factor is present (see Box 53.1). Imaging studies and cystoscopy are unnecessary in a male who has had a single UTI with no obvious complicating factors and whose infection responds promptly to treatment.

SELF-ASSESSMENT QUESTIONS

1. Which of the following antimicrobials is *not* recommended for first-line use for uncomplicated cystitis by the Infectious Disease Society of America treatment guidelines?
   A. Fosfomycin
   B. Ciprofloxacin
   C. Trimethoprim-sulfamethoxazole
   D. Nitrofurantoin
2. What approach has not been shown to be effective in the management of recurrent cystitis in females?
   A. Postcoital antimicrobial prophylaxis
   B. Daily antimicrobial prophylaxis
   C. Intermittent self-treatment
   D. Periodic screening and treatment of asymptomatic bacteriuria
3. Which of the following antimicrobials should be *avoided* in the treatment of prostatitis?
   A. Ciprofloxacin
   B. Nitrofurantoin
   C. Levofloxacin
   D. Trimethoprim-sulfamethoxazole
4. In which of the following conditions is screening for and treatment of asymptomatic bacteriuria indicated?
   A. Elderly males and females
   B. Catheterized males and females
   C. Pregnant patients and patients undergoing urologic instrumentation
   D. After treatment for acute pyelonephritis

REFERENCES

1. Nicolle LE. A practical guide to the management of complicated urinary tract infection. *Drugs*. 1997;53(4):583–592.
2. Hooton TM, Scholes D, Hughes JP, et al. A prospective study of risk factors for symptomatic urinary tract infection in young women. *N Engl J Med*. 1996;335(7):468–474.
3. Hooton TM, Stamm WE. Diagnosis and treatment of uncomplicated urinary tract infection. *Infect Dis Clin North Am*. 1997;11(3):551–581.
4. Scholes D, Hooton TM, Roberts PL, Gupta K, Stapleton AE, Stamm WE. Risk factors associated with acute pyelonephritis in healthy women. *Ann Intern Med*. 2005;142(1):20–27.
5. Foxman B, Barlow R, D'Arcy H, Gillespie B, Sobel JD. Urinary tract infection: self-reported incidence and associated costs. *Ann Epidemiol*. 2000;10(8):509–515.
6. Stamm WE, Hooton TM. Management of urinary tract infections in adults. *N Engl J Med*. 1993;329(18):1328–1334.
7. Nicolle LE, Gupta K, Bradley SF, et al. Clinical practice guideline for the management of asymptomatic bacteriuria: 2019 update by the infectious diseases Society of America. *Clin Infect Dis*. 2019;68(10):1611–1615.
8. Sobel JD. Pathogenesis of urinary tract infection. Role of host defenses. *Infect Dis Clin North Am*. 1997;11(3):531–549.
9. Johnson JR. Microbial virulence determinants and the pathogenesis of urinary tract infection. *Infect Dis Clin North Am*. 2003;17(2):261–278 (viii).
10. Lundstedt AC, McCarthy S, Gustafsson MC, et al. A genetic basis of susceptibility to acute pyelonephritis. *PLoS One*. 2007;2(9):e825.
11. Flores-Mireles AL, Walker JN, Caparon M, Hultgren SJ. Urinary tract infections: epidemiology, mechanisms of infection and treatment options. *Nat Rev Microbiol*. 2015;13(5):269–284.
12. Patterson JE, Andriole VT. Bacterial urinary tract infections in diabetes. *Infect Dis Clin North Am*. 1997;11(3):735–750.
13. Hooton TM. Clinical practice. Uncomplicated urinary tract infection. *N Engl J Med*. 2012;366(11):1028–1037.
14. Stamm WE, Counts GW, Running KR, Fihn S, Turck M, Holmes KK. Diagnosis of coliform infection in acutely dysuric women. *N Engl J Med*. 1982;307(8):463–468.
15. Hooton TM, Roberts PL, Cox ME, Stapleton AE. Voided midstream urine culture and acute cystitis in premenopausal women. *N Engl J Med*. 2013;369(20):1883–1891.
16. Bent S, Nallamothu BK, Simel DL, Fihn SD, Saint S. Does this woman have an acute uncomplicated urinary tract infection? *JAMA*. 2002;287(20):2701–2710.
17. Zhanel GG, Hisanaga TL, Laing NM, et al. Antibiotic resistance in Escherichia coli outpatient urinary isolates: final results from the North American urinary tract infection Collaborative Alliance (NAUTICA). *Int J Antimicrob Agents*. 2006;27(6):468–475.
18. Schito GC, Naber KG, Botto H, et al. The ARESC study: an international survey on the antimicrobial resistance of pathogens involved in uncomplicated urinary tract infections. *Int J Antimicrob Agents*. 2009;34(5):407–413.
19. Karlowsky JA, Lagace-Wiens PR, Simner PJ, et al. Antimicrobial resistance in urinary tract pathogens in Canada from 2007 to 2009: CANWARD surveillance study. *Antimicrob Agents Chemother*. 2011;55(7):3169–3175.
20. Manges AR, Johnson JR. Food-borne Origins of Escherichia coli causing extraintestinal infections. Clin Infect Dis.
21. Gupta K, Hooton TM, Naber KG, et al. International clinical practice guidelines for the treatment of acute uncomplicated cystitis and pyelonephritis in women: a 2010 update by the Infectious Diseases Society of America and the European Society for Microbiology and Infectious Diseases. *Clin Infect Dis*. 2011;52(5):e103–120.
22. Sanchez GV, Master RN, Karlowsky JA, Bordon JM. In vitro antimicrobial resistance of urinary Escherichia coli isolates among U.S. outpatients from 2000 to 2010. *Antimicrob Agents Chemother*. 2012;56(4):2181–2183.
23. Gupta K, Hooton TM, Roberts PL, Stamm WE. Short-course nitrofurantoin for the treatment of acute uncomplicated cystitis in women. *Arch Intern Med*. 2007;167(20):2207–2212.
24. Rodriguez-Bano J, Alcala JC, Cisneros JM, et al. Community infections caused by extended-spectrum beta-lactamase-producing Escherichia coli. *Arch Intern Med*. 2008;168(17):1897–1902.
25. Pullukcu H, Tasbakan M, Sipahi OR, Yamazhan T, Aydemir S, Ulusoy S. Fosfomycin in the treatment of extended spectrum beta-lactamase-producing Escherichia coli-related lower urinary tract infections. *Int J Antimicrob Agents*. 2007;29(1):62–65.
26. Dewar S, Reed LC, Koerner RJ. Emerging clinical role of pivmecillinam in the treatment of urinary tract infection in the context of multidrug-resistant bacteria. *J Antimicrob Chemother*. 2014;69(2):303–308.
27. U.S. Food and Drug Administration. Drug safety communication: FDA advises restricting fluoroquinolone antibiotic use for certain uncomplicated infections; warns about disabling side effects that can occur together. www.fda.gov/Drugs/DrugSafety/ucm500143.htm. Accessed 3/14/2021.
28. Hooton TM, Roberts PL, Stapleton AE. Cefpodoxime vs ciprofloxacin for short-course treatment of acute uncomplicated cystitis: a randomized trial. *JAMA*. 2012;307(6):583–589.
29. Hooton TM, Scholes D, Gupta K, Stapleton AE, Roberts PL, Stamm WE. Amoxicillin-clavulanate vs ciprofloxacin for the treatment of uncomplicated cystitis in women: a randomized trial. *JAMA*. 2005;293(8):949–955.
30. Stapleton A, Stamm WE. Prevention of urinary tract infection. *Infect Dis Clin North Am*. 1997;11(3):719–733.
31. Anderson GG, Dodson KW, Hooton TM, Hultgren SJ. Intracellular bacterial communities of uropathogenic Escherichia coli in urinary tract pathogenesis. *Trends Microbiol*. 2004;12(9):424–430.
32. Rosen DA, Hooton TM, Stamm WE, Humphrey PA, Hultgren SJ. Detection of intracellular bacterial communities in human urinary tract infection. *PLoS Med*. 2007;4(12):e329.
33. Talan DA, Stamm WE, Hooton TM, et al. Comparison of ciprofloxacin (7 days) and trimethoprim-sulfamethoxazole (14 days) for acute uncomplicated pyelonephritis pyelonephritis in women: a randomized trial. *JAMA*. 2000;283(12):1583–1590.
34. Hooton TM, Vecchio M, Iroz A, et al. Effect of increased daily Water intake in premenopausal women with recurrent urinary tract infections: a randomized clinical trial. *JAMA Intern Med*. 2018;178(11):1509–1515.
35. Kranjcec B, Papes D, Altarac S. D-mannose powder for prophylaxis of recurrent urinary tract infections in women: a randomized clinical trial. *World J Urol*. 2014;32(1):79–84.
36. Barbosa-Cesnik C, Brown MB, Buxton M, Zhang L, DeBusscher J, Foxman B. Cranberry juice fails to prevent recurrent urinary tract infection: results from a randomized placebo-controlled trial. *Clin Infect Dis*. 2011;52(1):23–30.
37. Nicolle LE. Uncomplicated urinary tract infection in adults including uncomplicated pyelonephritis. *Urol Clin North Am*. 2008;35(1):1–12.
38. Rahn DD, Carberry C, Sanses TV, et al. Vaginal estrogen for genitourinary syndrome of menopause: a systematic review. *Obstet Gynecol*. 2014;124(6):1147–1156.
39. Johnson JR, Russo TA. Acute pyelonephritis in adults. *N Engl J Med*. 2018;378(12):1162.
40. Warren JW, Abrutyn E, Hebel JR, Johnson JR, Schaeffer AJ, Stamm WE. Guidelines for antimicrobial treatment of uncomplicated acute bacterial cystitis and acute pyelonephritis in women. Infectious Diseases Society of America (IDSA). *Clin Infect Dis*. 1999;29(4):745–758.
41. Talan DA, Takhar SS, Krishnadasan A, et al. Fluoroquinolone-resistant and extended-spectrum beta-lactamase-producing Escherichia coli infections in patients with pyelonephritis, United States(1). *Emerg Infect Dis*. 2016;22(9).
42. Ahmed H, Farewell D, Jones HM, Francis NA, Paranjothy S, Butler CC. Incidence and antibiotic prescribing for clinically diagnosed urinary tract infection in older adults in UK primary care, 2004-2014. *PLoS One*. 2018;13(1):e0190521.
43. Drekonja DM, Rector TS, Cutting A, Johnson JR. Urinary tract infection in male veterans: treatment patterns and outcomes. *JAMA Intern Med*. 173(1):62-68.
44. Germanos GJ, Trautner BW, Zoorob RJ, et al. No clinical benefit to treating male urinary tract infection longer than seven days: an outpatient Database study. *Open Forum Infect Dis*. 2019;6(6):ofz216.

44a. Drekonja DM, Trautner B, Amundson C, Kuskowski M, Johnson JR Effect of 7 vs 14 days of antibiotic therapy on resolution of symptoms among afebrile men with urinary tract infection: a randomized clinical trial. *JAMA*. 2021;326(4):324–331.

45. Marsh KJ, Mundy L, Holter JJ, Johnson JR. Analysis of urine-specific antibiograms from veterans to guide empiric therapy for suspected urinary tract infection. *Diagn Microbiol Infect Dis*. 2019;95(4):114874.
46. Montelin H, Forsman KJ, Tangden T. Retrospective evaluation of nitrofurantoin and pivmecillinam for the treatment of lower urinary tract infections in men. *PLoS One*. 2019;14(1):e0211098.
47. Peterson J, Kaul S, Khashab M, Fisher AC, Kahn JB. A double-blind, randomized comparison of levofloxacin 750 mg once-daily for five days with ciprofloxacin 400/500 mg twice-daily for 10 days for the treatment of complicated urinary tract infections and acute pyelonephritis. *Urology*. 2008;71(1):17–22.
48. Funfstuck R, Ott U, Naber KG. The interaction of urinary tract infection and renal insufficiency. *Int J Antimicrob Agents*. 2006;28(suppl 1):S72–S77.
49. Gilbert DN. Urinary tract infections in patients with chronic renal insufficiency. *Clin J Am Soc Nephrol*. 2006;1(2):327–331.
50. Hooton TM, Bradley SF, Cardenas DD, et al. Diagnosis, prevention, and treatment of catheter-associated urinary tract infection in adults: 2009 international clinical practice guidelines from the infectious diseases Society of America. *Clin Infect Dis*. 2010;50(5):625–663.
51. Tambyah PA, Oon J. Catheter-associated urinary tract infection. *Curr Opin Infect Dis*. 2012;25(4):365–370.
52. Saint S, Greene MT, Krein SL, et al. A Program to prevent catheter-associated urinary tract infection in acute care. *N Engl J Med*. 2016;374(22):2111–2119.
53. Niel-Weise BS, van den Broek PJ. Urinary catheter policies for short-term bladder drainage in adults. *Cochrane Database Syst Rev*. 2005;3:CD004203.
54. Saint S, Kaufman SR, Rogers MA, Baker PD, Ossenkop K, Lipsky BA. Condom versus indwelling urinary catheters: a randomized trial. *J Am Geriatr Soc*. 2006;54(7):1055–1061.
55. Cardenas DD, Hooton TM. Urinary tract infection in persons with spinal cord injury. *Arch Phys Med Rehabil*. 1995;76(3):272–280.
56. Li L, Ye W, Ruan H, Yang B, Zhang S, Li L. Impact of hydrophilic catheters on urinary tract infections in people with spinal cord injury: systematic review and meta-analysis of randomized controlled trials. *Arch Phys Med Rehabil*. 2013;94(4):782–787.
57. Lipsky BA, Byren I, Hoey CT. Treatment of bacterial prostatitis. *Clin Infect Dis*. 2010;50(12):1641–1652.
58. Dembry LM, Andriole VT. Renal and perirenal abscesses. *Infect Dis Clin North Am*. 1997;11(3):663–680.
59. McHugh TP, Albanna SE, Stewart NJ. Bilateral emphysematous pyelonephritis. *Am J Emerg Med*. 1998;16(2):166–169.
60. Dobyan DC, Truong LD, Eknoyan G. Renal malacoplakia reappraised. *Am J Kidney Dis*. 1993;22(2):243–252.
61. Li L, Parwani AV. Xanthogranulomatous pyelonephritis. *Arch Pathol Lab Med*. 2011;135(5):671–674.
62. Cai T, Mazzoli S, Mondaini N, et al. The role of asymptomatic bacteriuria in young women with recurrent urinary tract infections: to treat or not to treat? *Clin Infect Dis*. 2012;55(6):771–777.
63. Dielubanza EJ, Schaeffer AJ. Urinary tract infections in women. *Med Clin North Am*. 2011;95(1):27–41.